LINUX

LINUX:

Linux is an free and Open-source operating system with high security. Linux is multi user based OS.

OS: Operating System

An Operating System (OS) is a software that acts as an interface between computer hardware components and the user.

Every computer system must have at least one operating system to run other programs.

Applications like Browsers, MS Office, Notepad Games, etc., need some environment to run and perform its tasks

TYPES OF OS

Key Features of Linux:

Linux follows the open-source model.

Linux has multiuser and multitasking capabilities.

Security features.

Linux is used for servers, embedded systems, supercomputers, etc

LINUX OS DISTRIBUTIONS:

Many of the users taken the linux OS and modified according to their requirements and released into the market with different names called Linux distribution.

RedHat

Ubuntu

Debian

Centos

Fedora

Opensuse

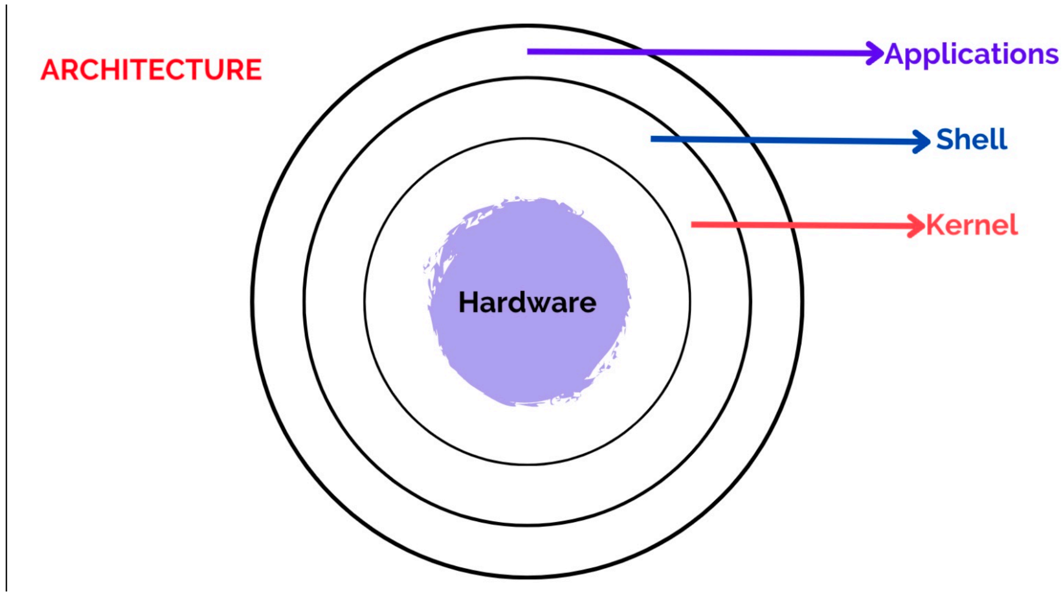
Kali Linux

Amazon Linux

Rocky Linux

HISTORY:

On Sep 17th 1991, Linus Torvalds a student at the university of Helsinki, Finland, He released the first version of the Linux kernel, known as Linux 0.01, as open-source software.
The Linux kernel is written in C language.
He wrote this program specially for his own PC
Firstly he wanted to name it as 'Freax' but later it became 'Linux'.
Today, supercomputers, smart phones, desktop, web servers, tablet, laptops and home appliances like washing machines, DVD players, routers, modems, cars, refrigerators, etc use Linux OS.



KERNEL:

It is the core or the heart of the operating system. It's the central part that manages and facilitates communication between the computer's hardware and software.

SHELL:

A shell that allows users to interact with the operating system. There are two types of shells.
Command Line Interface (CLI): Executes the command provided by user given in the form command and display the output in terminal.
Executes the process provided by user in graphical way and output is displayed in the graphical window

COMMAND:

It is an instruction/request given to the operating system by a user.
It tells computer to perform a particular task.

TERMINAL:

It is a text-based interface that allows you to interact with the operating system by typing commands.
It's a way for you to communicate with the linux machines

PRACTICAL SESSION:

By default we are in ec2-user, but if we want to perform any action we should be in root user because, root is the ultimate king of linux, root has full permissions, so that we can run any command anywhere.

To login as root user : sudo -i (or) sudo su -
Here sudo : super user do
to logout from root user and go back to ec2-user : exit

In Linux we have different types of commands

1. SYSTEM COMMANDS
2. HARDWARE COMMANDS
3. FILE COMMANDS
4. PERMISSION COMMANDS
5. USER COMMANDS
6. SEARCH COMMANDS
7. NETWORKING COMMANDS

SYSTEM COMMANDS:

Used to get system information
uname : used to get type of OS
uname -r : used to get kernel version of our OS
uname -a : used to get full info about OS
clear: this command is used to clear the clear (or) you can use ctrl + l as a short cut
uptime : used to get since how long our system is in running state
uptime -p : this will give only time
hostname: used to get private dns name of our system
hostname -i : used to get private ip of our system
hostnamectl set-hostname “swiggy” : used to change hostname
ip addr : used to get private IP
ip route : used to get private IP
ifconfig : used to get private IP
date : to get todays date
timedatectl : used to get timezones
timedatectl set-timezone Asia/Kolkata : used to change Timezone to IST
who : used to see how many users have been loggin into your system
whoami : used to see the current user.

HARDWARE COMMANDS:

These are the hardware commands
lscpu : cpu information of our servertop : used to get cpu utilization
df -h : used to get the 8 volume details (used, available)
free : used to get the free ram in our server
free -m : used to convert the ram in MB
free -h : used to ram info in human readable format

FILE COMMANDS:
touch filename : used to create a file
touch aws azure gcp : used to create multiple files
touch linux{1..5} : this command will create 5 files (linux1,inux2, linux3 .... linux5)
rm filename : remove file with permissions (rm means remove)
rm -f filename : remove file without permissions (-f means forcefully)
rm -f aws azure gpc : remove multiple files without permissions
rm -f linux{1..5} : this command will remove linux files from 1 to 5 without permissions
rm -f * : used to delete all files
rm -f *.txt : used to delete all files with .txt extension
rm -f a* : this command will delete all files which are stared with â€œaâ€ letter

FOLDER COMMANDS:
mkdir foldername : used to create a folder
mkdir git maven jenkins : used to create multiple folders
mkdir docker{1..5} : used to create 5 folders at the same time
rmdir foldername : used to remove empty folder
rmdir git maven jenkins : will remove multiple empty folders
rmdir docker{1..5} : used to remove 5 docker empty folders
rmdir * : used to remove all empty folder
rm -rf * : used to remove all files & folders and also it is used to remove non empty folders

LIST THE FILES:
ll : used to get list of files
ls : used to get list of files

ll vs ls

ll : will give the full info about files/folders
ls : it will give only file/folder names

CHANGE DIRECTORY:

cd foldername : used to change directory
cd : used to go to root directory
cd - : used to go to previous folder
cd ../ : used to go to one folder back
cd ../../ : used to go to 2 folders back

DIRECTORY COMMANDS:

mkdir folder1/folder2 : this will creates folder2 inside folder1
ll folder1 : used to get list of files & folder which are present in folder1
touch folder1/aws.txt : used to create file inside a folder
mkdir -p aws/azure/gcp/ccit : used to create parenting folder (folder inside the folder) automatically

TIP OF THE DAY:

use tab to get the full file name or folder name
use upper-arrow to get previously executed commands
use ctrl + u to delete the entire command
use ctrl + e to go to end of the command
use ctrl + a to go to starting of the command

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ll : used to see the list of files in order (A-Z)
ll-t : used to see the list of the files based on modification/creation time
lll-r : used to see the files in reverse order (Z to A)
ll -a : used to see all files including hidden

CAT COMMAND:

cat command is used to read the data from a file, it is also used to append the data in a file
cat filename: used to read the data from a file
cat -n filename : used to read the data including line numbers of a file
cat>filename : used to overwrite the data
cat>>filename: used to append the data
head filename : used to print first 10 lines of a file
tail filename : used to print last 10 lines of a file
head -5 filename : used to print first 5 lines of a file
tail -7 filename : used to print last 7 lines of a file
sed -n '3,15p' filename : used to print lines from 3 to 15
sed -n '16p' filename : used to print 16 th line of a file

CP COMMAND: <cp source destination>

cp command is used to copy the files or data from one place to another place
ex: cp file1 file2

By the above command, the data from file1 copies into file2. But the problem is it will overwrite the data which are present in file2.

To overcome this issue we will use cat command.
cat source_file (file1) >> destination_file (file2)

MV COMMAND: <mv source destination>

mv command is used to move the files or data from one place to another place. This is also used to rename the files.
ex: mv Jenkinsfile Folder-1

NOTE:

Cat command is used to append the data, but here the problem is, it is not possible to modify the data. To avoid this issue we can use editor in linux.
There are 2 types of editors in linux

1. Vim editor
2. Nano editor

VIM EDITOR:

It is used to edit the files in linux machines, It has 3 modes

1. Command mode
2. Insert mode
3. Save & quit mode

To open any file in vim editor : vim filename (or) vi filename

1. COMMAND MODE:

This is the default mode in vim editor, It is used to perform some actions like used to copy the data, delete the data and we can make undo and redo the changes as well.

gg : used to go to 1st line of the file
G : used to go to last line of the fileM : used to go to middle of the file
4gg : used to go to 4th line of the file
17gg : used to go to 17th line of the file
:23 : used to go to 23rd line of the file
:set number : used to set numbers of the file
yy : used to copy the line
4yy : used to copy 4 lines from our cursor
p : paste the copied content
10p : paste the copied content 10 times
dd : used to delete the line
5dd : used to delete 5 lines from the cursor
u : used to undo the changes
crtl + r : used to redo the changes.
/word : used to search for a word in a file
?word : used to search for a word in a file
:%s/old/new/ : used to replace

2. INSERT MODE:

This mode is used to insert the data or make any modifications in a file.
To go to insert mode : i
To go back to command mode : esc
To go to the ending of the line : A
To go to starting of this line : I
To create a new line above the cursor : O
To create a new line below the cursor : o

DIFFERENCE B/W COMMAND MODE KEYS & INSERT MODE.

If we perform command mode keys we will be in command mode only
If we perform insert mode keys we will go to insert mode.

3. SAVE & QUIT MODE:

This is used to save the data and quit from vim editor
To save the data - :w
To quit from vim editor - :q
To quit forcefully - :q!
To save & quit at a time - :wq
To save & quit forcefully at a time - :wq!

WC COMMAND:

wc filename : used to get no of words, letters and lines of a file

wc -w filename : used to get no of words

wc -c filename : used to get no of characters

wc -l filename : used to get no of lines

PERMISSION COMMANDS: -

rw-r--r--. 1 root root 0 Oct 27 15:24 amazon

(-) : type of the file

rw-r--r-- : PERMISSIONS OF A FILE

r = read = 4

w = write = 2

x = execute = 1

- = nothing = 0

USER = rw- ------> 4+2+0 = 6

GROUP = r-- ------> 4+0+0 = 4

OTHERS = r-- -----> 4+0+0 = 4

644

- to change the permissions of a file : chmod 754 filename
- to change permissions to multiple files : chmod 777 file1 file2 file3
- to change permissions to folder only : chmod 654 folder
- to change permissions to files only in folder : chmod 567 folder/*
- to change permissions to file & folder at a time : chmod -R 123 folder

USER & GROUP COMMANDS

USER COMMANDS

To check the list of users : cat /etc/passwd
To create a user : useradd ravi
ravi:x:1001:1001::/home/ravi:/bin/bash
ravi -----> username
x -----> it stores the users password
1001 ----> UID (User ID)
1001 -----> GID (Group ID)
/home/ravi ---> path of the user
/bin/bash -----> stores the commands
logout from the current user: exit
to set a password to user : passwd username
To get UID & GID id a user : id username
To delete a user : userdel username
To delete user along with folder : userdel -r username
To delete user forcefully : userdel -f username
To set expiry date to user : useradd -e <date> username
To create user withour folder : useradd -M username

GROUP COMMANDS

To create. a group : groupadd groupname
To list of groups : cat /etc/group
To delete a group : groupdel groupname
To attach a user to a group : usermod -a -G groupname username

CHANGING OWERNS OF A FILE

To change user of a file : chown newuser filename
To change group of a file : chgrp newgroup filename
To change user & group at a time : chown username:groupname filename
To change user & group to multiple files : chown username:groupname f1 f2 f3
To change owners of folder : chown user:group foldername
To change owners of folder along with the files present in folder : chown -R user:group foldername
To change owners of files that are present in folder : chown user:group foldername/*

SEARCH COMMANDS:

It is used to search for a file, To search for a file we have 2 commands

1. Grep Command
2. find command
3. locate Command

1. GREP (Global Regular Expression Print):

It is used to search for a word in a file

SYNTAX: grep "word" filename

- grep "word" filename : used to search for a word in a file
- grep -n "word" filename : it prints the data along with line numbers
- grep -c "word" filename : it prints no of occurances of a word
- grep -i "word" filename : used to search for a case-sensitive

2. FIND COMMANDS:

SYNTAX: find path -name file_name

- •find . -name file : used to find a file in current directory
- •find /proc/ -name filename : used to find a file in proc directory
- •find . -type d -name folder : used to find a folder in current directory
- •find . -type f -name <file1.txt> : used to find a file in current directory
- •find . -type f -perm 777 : Finds all the files whose permissions are 777 in the current directory
- •find . -type f ! -perm 777 : Finds all the files whose permissions are NOT 777 in the current directory
- •find / -user <username> : Finds all the files specific user owned in / directory
- •find / -group groupname : Finds all the files specific group owned in / directory

3. LOCATE COMMANDS:

SYNTAX: locate filename

NOTE:

If you are using locate command, we have to update the db of linux before performing locate command.
find vs locate : find commands will search as per the path that we mentioned in command, but locate command will search for a file on entire database.